DC-URS — engineering review and adoption plan

From: Engineering
Date: 2026-07-27
Subject: Review of the Delta Climate Urban Resilience Score, and what it takes to ship it

---

Summary

We're adopting it, and it replaces the Green Score. I've read the specification, run the reference implementation, stress-tested the formulation, and checked the anchor citation. This is a better instrument than what it replaces, for two reasons beyond the ones in your document.

It shrinks our biggest exposure. We spent last week measuring how accurate our heat model actually is: **± 3.5 °C at night, ± 5.0 °C by day**, and the daytime figure cannot be improved without better weather data. (Since this note was first issued we have also measured the *spatial* skill — whether the map places heat correctly *within* a ward — and it is weak: r = 0.11, below the r = 0.23 of a plain vegetation map. That is disclosed in the product and detailed in §5A of the methodology. It strengthens rather than weakens the argument below: DC-URS scores on ward-level indicators and never touches the per-cell field, so the finding does not reach the index at all.) In the Green Score, modelled cooling carried two-thirds of the weight — so that uncertainty went almost straight through to the headline number. In DC-URS, heat carries `w_H = 0.25` and is normalised over a 20 K span, so the same error moves the score by about **5 points out of 100**, a quarter of one tier band. Your index makes our known weakness much less load-bearing. That is a strong argument for it on its own.

It answers the objection already on the table. The standing criticism of the Green Score is that it applies Berlin's Biotope Area Factor — a temperate, European, site-level standard — to whole wards in tropical Kolkata. Moving to the IPCC AR6 hazard / exposure / vulnerability triad fixes that at the root. Better still, there is now Indian precedent underneath it (§5), so this becomes a genuinely local instrument rather than a differently-imported one.

The rest of this note is eight things to correct before it ships, and what replacement means in practice.

---

1. The one structural change I'd ask for

The score adds its three pillars; IPCC risk multiplies them.

In AR6, risk is *conjunctive*: if exposure is near zero, risk is near zero however severe the hazard. A weighted sum lets a strong pillar buy off a weak one. Worked example:

Ward	Thermal hazard	Adaptive capacity	DC-URS
A — dangerous heat, beautifully green	0.85	0.90	**57.2**
B — cool, bare	0.30	0.35	**49.0**

Ward A is 0.55 hotter and scores **8.3 points better**. Greenery is buying its way out of lethal heat, and both land in the same action tier.

The fix is nearly free — multiply instead of add:

```
DC-URS = 100 × ACI^0.40 × (1 − EVI)^0.35 × (1 − THI)^0.25
```

This flips the ordering above, and on **your own two worked wards it barely moves**: Ballygunge 20.0 → 19.8, Baruipur 65.2 → 64.5. Your calibration survives; the compensation loophole doesn't.

---

2. Three defects in the greenness pillar

NDVI and FVC are the same variable. Your §3 defines `FVC = (NDVI − NDVI_bare)/(NDVI_veg − NDVI_bare)` — a straight-line rescaling of NDVI. So `φ₁·NDVI + φ₂·FVC` spends **0.80 of the greenness weight on one input counted twice**. Suggest keeping one and spending the freed weight on something independent, such as tree-canopy fraction.

The stability index scores water as perfect vegetation. `VSI = 1 − σ/mean` with water's NDVI ≈ −0.30 makes the ratio negative; the `clip(…, 0, 1)` floors it at zero, so **VSI = 1.00 — maximum vegetation stability, for a river**. Bare ground gets 0.00. The Hooghly runs through our study area, so this fires immediately.

Stability cannot be simulated. VSI measures multi-year persistence. A tree planted today has no history: honest values give newly planted cover VSI ≈ 0.27 against a mature canopy's 0.95 — so the simulator would show **planting trees making the score worse**. Options are to freeze it in scenario mode, or drop it. Given it is also the broken term, I lean toward dropping it and carrying persistence differently.

---

3. Three inconsistencies to reconcile

The weights table and the code disagree.

Pillar	§4 table implies	§5 code uses	
Exposure	0.429 / 0.286 / 0.286	0.45 / 0.30 / 0.25	mismatch
Adaptive capacity	0.50 / 0.25 / 0.25	0.50 / 0.30 / 0.20	mismatch
Hazard	0.40 / 0.40 / 0.20	0.40 / 0.40 / 0.20	agrees

Two different normalisation methods. §3 specifies min-max scaling; §5's code uses fixed divisors. These behave very differently, and **min-max would be disqualifying for a product**: with three wards the best always scores 1.0 and the worst 0.0, and adding a fourth ward silently changes every existing score. The code's fixed anchors are the right choice — the text should follow the code, with each anchor justified.

The hazard term goes blind where it matters most. The daytime term saturates at 45 °C, so Ballygunge at 42.5 °C and at 47.5 °C score identically. Heat action plans exist precisely for the wards above that line.

---

4. On the Czekajlo citation

The paper is real and well-regarded — Czekajlo, Coops, Wulder et al., *International Journal of Applied Earth Observation and Geoinformation* 93:102210 (2020), 59 citations. I read it.

What it actually does is derive greenness by **spectral unmixing of annual Landsat composites across 18 Canadian cities, 1984–2016**, and its score characterises *current level together with trajectory*. It is not `NDVI + FVC + VSI`. The concept we're borrowing is sound and worth crediting, but the formula in the specification isn't theirs, and describing it as their metric would be a problem the first time a technical reviewer opens the paper. We were caught by a borrowed benchmark once already this month, so I'd rather be careful here than quick.

Two practical notes follow from actually reading it. It is calibrated on Canadian cities, so it carries the same transplant caveat we are leaving Berlin over — the method transfers, the parameters do not. And it uses 33 years of annual composites for a reason: a single-scene NDVI in a monsoon city swings enormously between wet and dry season, which is the same seasonal signal we measured in the satellite work. Our NDVI inputs need to be seasonal composites, not one date.

---

5. The best news: there is Indian precedent, and a way to check ourselves

The socio-economic pillar looked like the hardest part to defend. It turns out to be the best-supported:

- **Rathi et al. (2021)**, 63 citations — heat vulnerability index across exposure, sensitivity and adaptive capacity for urbanites of four Indian cities, **including Kolkata**.
- **Azhar et al. (2017)**, 126 citations — heat-wave vulnerability mapping for India from Census of India 2011 and the District Level Household Survey.

- **Sharma (2026)** — ward-level heat vulnerability across 80 wards of Jodhpur.

These give us both a construction method with Indian precedent *and* an independent Kolkata ward-level map to rank-correlate our own output against. That is a real validation route for a composite index, which normally has no ground truth at all. It is also the "local, not European" claim actually delivered rather than asserted.

The honest limitation: the demographic inputs rest on **Census of India 2011**, because the 2021 census was postponed. That is fifteen-year-old data carrying 0.10 of the score. We should state that on the face of the tool rather than have a client find it.

---

6. What "replaces the Green Score" means in practice

It does not cost us the simulator. The Green Score already worked as *baseline → intervention → new score*, and DC-URS works the same way: one score, evaluated in two states.

What changes is that part of it cannot move. Population density, built density and social vulnerability don't respond to any intervention, so **35 % of the score is fixed per ward**. Run the numbers on Ballygunge: with a *physically perfect* retrofit — full canopy, 0.60 albedo, a cooling refuge next door, surface temperature down to the rural baseline — the best it can reach is **61.4**. It can never leave "Moderate Resilience".

I want to argue that this is the most valuable output the tool has, not a limitation:

> *28 of Ballygunge's missing points are demographic exposure. Trees cannot fix them. That needs housing, health outreach and cooling-centre policy.*

No greenness score can produce that sentence. I'd make the immovable portion **visible by design** — it separates what greening can achieve from what only policy can, which is exactly the distinction a municipality is paying us to draw.

---

7. Two things I need from you

The socio-economic inputs. Your worked example puts Ballygunge at 7.2 and Baruipur at 4.1 on social vulnerability. Ballygunge is one of the more affluent parts of south Kolkata, so I suspect those are placeholders rather than intended values. I can derive real figures from Census 2011 using the Rathi/Azhar method — I'd like your sign-off on that approach before I build it.

There's a complication worth knowing early: **our three study areas sit under three different local bodies.** Ballygunge is KMC Ward 68; Baruipur and Barrackpore are separate municipalities with their own boundaries and returns. We need one consistent spatial unit or the exposure pillar isn't comparable across the three.

A judgement call on the public tier labels. The classification prints "Critical Hotspot — extreme emergency risk" for scores under 40. On his own figures, **Ballygunge scores 20.0** and would carry that label on a public page, by name. That is a strong public statement about a real neighbourhood, and residents, councillors and the corporation will all read it. I'm not saying it's wrong — it may be exactly the message. But it's a positioning decision rather than an engineering one, and I'd rather raise it now than after publication.

---

8. Proposed sequence

Phase	Work
0	Settle the spatial unit across the three local bodies. Nothing downstream is meaningful without it.
1	Observed pillars — greenness, albedo, built density, refuge access — from satellite and OSM. No modelling.
2	Exposure pillar from Census 2011, built the Rathi/Azhar way, vintage disclosed.
3	Validation — rank correlation against published Kolkata vulnerability, sensitivity analysis, and an uncertainty band carried through from our measured ± 3.5 / ± 5.0 °C.
4	Scenario layer over the pillars that can honestly move, with the fixed portion shown explicitly.

Full formulas and every constant will land in `docs/dc-urs-spec.md` before implementation starts.

Sources

Source	Used for	Link
Czekajlo, Coops, Wulder et al. (2020), *IJAEO* 93:102210	Multi-decadal greenness scoring; what it does and does not define	doi:10.1016/j.jag.2020.102210
Rathi, Chakraborty, Mishra & Dutta (2021), *Int. J. Environ. Res. Public Health*	Heat vulnerability index for urbanites of four Indian cities, including Kolkata	doi:10.3390/ijerph19010283
Azhar, Saha, Ganguly & Mavalankar (2017), *Int. J. Environ. Res. Public Health*	Heat-wave vulnerability mapping for India from Census 2011 + DLHS	doi:10.3390/ijerph14040357
IPCC AR6 WGII	Risk as hazard × exposure × vulnerability	ipcc.ch/report/ar6/wg2
Delta measured accuracy, ward level	± 3.5 °C night / ± 5.0 °C day, 49 ECOSTRESS scenes	`docs/green-score-methodology.md` §5A
Delta measured accuracy, within a ward	r = 0.11 against ECOSTRESS at 70 m, below the r = 0.23 of a vegetation map — pattern **not** validated	`docs/green-score-methodology.md` §5A

Every DOI above was resolved against Crossref before this was sent. Written to be argued with — if any of the eight points is wrong, I'd rather hear it from you than from a client.